JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Computer Science & Engineering
"A Place to Learn, A Chance to Grow"
Session: 2025-26

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Question Bank

Branch: - Computer Science & Engineering **Semester/Year: - V Sem / 3rd Year**

Subject: - Machine Learning **Subject Code: - CS5T002**

Subject In-charge: -Prof. Aachal Wani

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UNIT- 01

02 Marks Questions

Q.1 Define Machine Learning.
Q.2 List out the different issues in Machine Learning.
Q.3 Draw the diagram for showing the steps involved in designing a learning system.
Q.4 Enlist the Advantages and Disadvantages of Unsupervised Learning.
Q.5 State the Well Posed Learning Problems with Examples.
Q.6 Write a Short note on Reinforcement Learning.
Q.7 Give the Advantages and Disadvantages of Supervised Learning.
Q.8 Rewrite the Concept of Semi supervised Learning.

04 Marks Questions

Q.1 Distinguish between Supervised and Unsupervised Learning.
Q.2 Examine Semi-Supervised Learning with Example and Advantages and Disadvantages.
Q.3 Express the Concept of Well Posed Learning Problems with Four Examples.
Q.4 Illustrate and List the Different Issues in Machine Learning.

07 Marks Questions

Q.1 Explain the Different Applications of machine Learning.
Q.2 Examine the Step's in Developing a Machine Learning.
Q.3 Illustrate the various real-world applications of Machine Learning.
Q.4 Describe complete Life Cycle of Machine Learning by describing each phase in detail.
Q.5 Analyse the Advantages & Disadvantages of the following

a) Supervised Learning.
b) Unsupervised Learning.
c) Reinforcement Learning.
d) Semi supervised Learning.

Q.6 Examine the steps involved in designing a learning system with showing the diagram.

12 Marks Questions

Q.1 Describe the Following About Machine Learning:

a) Applications of Machine Learning
b) Life Cycle of Machine Learning

Q.2 Analyze the concept of Machine Learning and evaluate its different types—supervised, unsupervised, Semi-supervised and reinforcement learning—with suitable real-world examples.

Q.3 Explain the concept of Machine Learning and evaluate its different types with suitable real-world examples.

UNIT- 02

02 Marks Questions

Q.1 Identify the Concept of K-mean Clustering.

Q.2 Illustrate the Applications of Clustering.

Q.3 Identify the Concept of Hierarchical Clustering.

Q.4 Classify the Different Applications of Clustering.

Q.5 State the Term Hierarchical Clustering with Example.

Q.6 Write a Short Note on Clustering.

04 Marks Questions

Q.1 Write short note on K-mean Clustering.

Q.2 Examine Different Applications of Clustering.

Q.3 Express the Step-by-Step Algorithm of Batchelor- Wilkins.

07 Marks Questions

Q.1 Explain the Concept of Hierarchical Agglomerative Clustering (HAC) and illustrate its workflow with a diagram.

Q.2Analyse about following Algorithms:

a) Batchlor- Wilkin's Algorithm
b) K-mean Clustering Algorithm.

Q.3 Explain Hierarchical Agglomerative Clustering (HAC) and also Illustrate the working process of Hierarchical Agglomerative Clustering (HAC) with the help of a suitable diagram.

12 Marks Questions

Q.1 Rewrite the following Concept with their Algorithms:

a) K-mean Clustering.
b) Batchlor- Wilkin's Algorithm
c) Decision Tree

UNIT- 03

02 Marks Questions

Q.1 From the deck of Cards, Find the Probability of the card being picked is King Given that it is the Face Card.

- This can be represented as: p(King | Face).

- There are total 52 cards in a desk (n-52), from Which 12 cards are Face Cards: King, Queen and jack with Club, Diamond, heart and Spade.
- There is a set of Face cards with 12 members.
- There is a Subset of King Cards with 4-members. So, 4 Face cards out of 12 Face cards (4/12) are King cards. Use Bayes Theorem to Calculate.

Q.2 Restate the concept of the Bayes Optimal Classifier and illustrate it using its formula.
Q.3 Restate the concept of the Bayes Theorem and illustrate it using its formula.
Q.4 Identify the concept of the Bayes Optimal Classifier and express it mathematically using the appropriate formula.
Q.5 Underline the Concept of Naïve Bayes Classifier.
Q.6 In a Certain Population, the Probability that a person has Dengue is P(D)=1/30000. Dengue Fever Cusses Neck Pain 80% of the time. The Probability that a person experiences neck pain from any cause is P(N)-0.02 what is the probability that a person has dengue given that they have neck pain? Use Bayes' theorem to calculate.

04 Marks Questions

Q.1 Illustrate the concept of the Bayes Theorem and illustrate it using its formula.
Q.2 For the Given Dataset, Apply Naïve –Bayes Algorithm and Predict the Outcome for a car, Car= {Red, Domestic, SUV}

Color	**Type**	**Origin**	**Stolen**
Red	Sports	Domestic	Yes
Red	Sports	Domestic	No
Red	Sports	Domestic	Yes
Yellow	Sports	Domestic	No
Yellow	Sports	Imported	Yes
Yellow	SUV	Imported	No
Yellow	SUV	Imported	Yes
Yellow	SUV	Domestic	No
Red	SUV	Imported	No
Red	Sports	Imported	Yes

Q.3 Write short note on Naïve Bayes Classifier.
Q.4 Given the Training data in the table (Buy Computer Data), Predict the Class of the Following New Example Using Bayes Optimal Classifier.
Buy Computer = {Age = <=30, income = medium, student = yes, Credit-Rating = fair}

Age	Income	Student	Credit Rating	Buy's Computer
<=30	High	No	Fair	No
<=30	High	No	Excellent	No
31…40	High	No	Fair	Yes
>=40	Medium	No	Fair	Yes
>=40	Low	Yes	Fair	Yes
>=40	Low	Yes	Excellent	No
31…40	Low	Yes	Excellent	Yes

<=30	Medium	No	Fair	No
<=30	Low	Yes	Fair	Yes
>=40	Medium	Yes	Fair	Yes
<=30	Medium	Yes	Excellent	Yes
31..40	Medium	No	Excellent	Yes
31..40	High	Yes	Fair	Yes
>=40	Medium	No	Excellent	No

07 Marks Questions

Q.1 Analyse the Following Concept and illustrate it using its formula.

a) Naïve Bayes Classifier.

b) Bayes Optimal Classifier

Q.2 Given the Training data in the table (Buy Computer Data), Predict the Class of the Following New Example Using Bayes Optimal Classifier.

Buy Computer = {Age = <=30, income = medium, student = yes, Credit-Rating = fair}

Age	Income	Student	Credit Rating	Buy's Computer
<=30	High	No	Fair	No
<=30	High	No	Excellent	No
31…40	High	No	Fair	Yes
>=40	Medium	No	Fair	Yes
>=40	Low	Yes	Fair	Yes
>=40	Low	Yes	Excellent	No
31…40	Low	Yes	Excellent	Yes
<=30	Medium	No	Fair	No
<=30	Low	Yes	Fair	Yes
>=40	Medium	Yes	Fair	Yes
<=30	Medium	Yes	Excellent	Yes
31..40	Medium	No	Excellent	Yes
31..40	High	Yes	Fair	Yes
>=40	Medium	No	Excellent	No

Q.3 Consider the given Dataset, Apply Naïve Bayes Algorithm and Predict that is a Fruit has the following Properties then which type of fruit it is.

Fruit = {yellow, Sweet, Long}

Frequency Table

Fruit	Yellow	Sweet	Long	Total
Mango	350	450	0	650
Banana	400	300	350	400
Other's	50	100	50	150
Total	800	850	400	1200

Q.4 Describe the concept of the Bayes Optimal Classifier and illustrate it using its formula and solve the Following Example: Consider the Given Dataset, Apply Bayes Optimal Classifier Algorithm and predict that if a play has Following Properties.
Play Tennis = {Outlook = Sunny, Temperature = hot, Humidity= Normal, Wind = Strong}

Day	Outlook	Temperature	Humidity	Wind	Play Tennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

12 Marks Questions

Q.1 Consider the Given Dataset, Apply Naïve Baye's Algorithm and Predict that if a Tennis Play has following Properties
Tennis Play = {Outlook=Sunny, Humidity=High, Wind= Strong}
Tennis Play = {Outlook=Sunny, Humidity=High, Wind= Weak}

Day	Outlook	Humidity	Wind	Play Tennis
D1	Sunny	High	Weak	No
D2	Sunny	High	Strong	No
D3	Overcast	High	Weak	Yes
D4	Rain	High	Weak	Yes
D5	Rain	Normal	Weak	Yes
D6	Rain	Normal	Strong	No
D7	Overcast	Normal	Strong	Yes
D8	Sunny	High	Weak	No
D9	Sunny	Normal	Weak	Yes
D10	Rain	Normal	Weak	Yes
D11	Sunny	Normal	Strong	Yes
D12	Overcast	High	Strong	Yes
D13	Overcast	Normal	Weak	Yes
D14	Rain	High	Strong	No

Q.2 Given the Training data in the table (Buy Computer Data), Predict the Class of the Following New Example Using Naïve Bayes Classifier and Bayes Optimal Classifier.
Buy Computer = {Age = <=30, income = medium, student = yes, Credit-Rating = fair}

Age	Income	Student	Credit Rating	Buy’s Computer
<=30	High	No	Fair	No
<=30	High	No	Excellent	No
31…40	High	No	Fair	Yes
>=40	Medium	No	Fair	Yes
>=40	Low	Yes	Fair	Yes
>=40	Low	Yes	Excellent	No
31…40	Low	Yes	Excellent	Yes
<=30	Medium	No	Fair	No
<=30	Low	Yes	Fair	Yes
>=40	Medium	Yes	Fair	Yes
<=30	Medium	Yes	Excellent	Yes
31..40	Medium	No	Excellent	Yes
31..40	High	Yes	Fair	Yes
>=40	Medium	No	Excellent	No

UNIT- 04

02 Marks Questions

Q.1 Define Linear Regression.
Q.2 Classify a Different Types of Linear Regression.
Q.3 Define term Regression with Example.
Q.4 Define Bayesian Linear Regression.

04 Marks Questions

Q.1 Discuss Linear Regression with their types.

07 Marks Questions

Q.1 Describe Regression in detail and also describe Linear Regression with their Types.

UNIT- 05

02 Marks Questions

Q.1 Rewrite the Following term a) Hypothesis Space b) Hypothesis Space Search
Q.2 Restate the Decision Tree Algorithm.
Q.3 Define the following terminology w.r.t. Decision Tree.
a) Root Node. b) Leaf Node. c) Splitting. d) Pruning
Q.4 Rewrite the term Hypothesis Space Search with Example.
Q.5 Write a Short Note on Decision Tree.
Q.6 State the Following term in the context of Decision Tree
a) Hypothesis Space b) Hypothesis Space Search
Q.7 Give the any of the Example of Hypothesis Space Search

04 Marks Questions

Q.1 Discuss the Hypothesis Space Search in Decision Tree in detail.

Q.2Discuss the following terminology w.r.t. Decision Tree.

1. Root Node.
2. Leaf Node.
3. Splitting.
4. Branch/Sub Tree.
5. Pruning
6. Parent/Child Node.

Q.3 Illustrate the Different Methods of Sampling.

Q.4 Discuss and List the Different Types of Sampling.

07 Marks Questions

Q.1 Illustrate in Detail Concept of Decision Tree and also Outline the main steps followed in constructing a Decision Tree algorithm.

Q.2 Examine in detail about Hypothesis Testing with their Types.

Q.3 Illustrate Basic Sampling Theory with their Step by Step Process.

Q.4 Analyse the Concept of Basic Sampling Theory with their Step by Step Process involve in creating one with a neat diagram.

Q.5 Discuss the concept of Hypothesis Testing by explaining its types, and graphical representation with suitable examples.

12 Marks Questions

Q.1 Rewrite the Concept of Basic Sampling Theory and also Rewrite following about Sampling:

a) Process of Sampling
b) Types of Sampling